

Predicting Airline Booking Completions with a Tuned Random Forest Classifier

Solving severe class imbalance in flight-booking behavioral data to identify high-intent customers — British Airways customer booking dataset (50,000 records)

1 THE PROBLEM

In online flight bookings, the vast majority of website visitors are "window shoppers" who search for flights but never purchase. A standard, out-of-the-box classifier trained on this data will default to predicting "No Booking" for nearly everyone — maximizing overall accuracy while missing almost every real buyer. For an airline, that means wasted marketing spend and lost opportunities to close sales on customers who were genuinely close to converting.

2 THE SOLUTION

I built a Random Forest Classifier pipeline specifically configured for severely imbalanced behavioral data. Rather than letting the model default to the majority class, I re-balanced the learning weights with `class_weight='balanced'` and tuned tree depth and split thresholds to sharpen the model's sensitivity to purchase-intent signals.

3 BUSINESS VALUE

- Automated precision marketing:** Trigger targeted incentives (baggage discounts, seat-upgrade reminders) toward window-shoppers showing elevated conversion potential.
- Demand & inventory planning:** Give revenue management visibility into which routes attract committed buyers versus casual browsers.
- Reduced customer friction:** Personalize the booking journey using signals like trip length and advance planning time.

4 TECHNICAL PIPELINE

- Class weighting** — `class_weight='balanced'` penalizes misclassified booking completions during tree construction.
- Hyperparameter tuning** — GridSearchCV (3-fold), optimizing `max_depth`, `min_samples_split`, `n_estimators`, scored on F1.
- Feature importance mapping** — Gini importance to identify the behavioral drivers of booking completion.

5 KEY RESULTS

59% RECALL — BUYER CLASS	68.25% OVERALL ACCURACY	~26% PRECISION — BUYER CLASS
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The tuned model catches more than half of actual bookers — a major improvement over a naive majority-class model, which would catch effectively none. Accuracy was deliberately traded down from a higher-but-useless "predict everyone as non-buyer" baseline in order to make the model actually useful for targeting.

6 TOP PREDICTIVE DRIVERS

RANK	FEATURE	BEHAVIORAL SIGNAL
1	flight_duration	Trip length is the strongest indicator of purchase completion
2	length_of_stay	Duration at destination heavily influences conversion likelihood
3	purchase_lead	How far in advance a customer plans reveals underlying intent

7 TRADE-OFFS & HONEST CAVEATS

Precision/recall trade-off: Optimizing for recall increases false positives (~26% precision on the buyer class). That's the right call for a low-cost marketing nudge, but would need re-tuning for a higher-cost intervention such as a sales call.

Fold-to-fold stability: A follow-up 5-fold cross-validation run showed noticeable F1 variance across folds, indicating performance depends somewhat on the specific train/test split. Next steps: SMOTE/undersampling, gradient boosting with `scale_pos_weight`, or additional behavioral features to stabilize results before production.

8 KEY TAKEAWAY

Data science isn't about chasing 99% accuracy on paper — it's about understanding the business context, correctly framing the cost of a missed buyer versus a wasted discount, and being transparent about where the model still needs work.

9 TECH STACK

Python Scikit-Learn Pandas NumPy Matplotlib Seaborn GridSearchCV